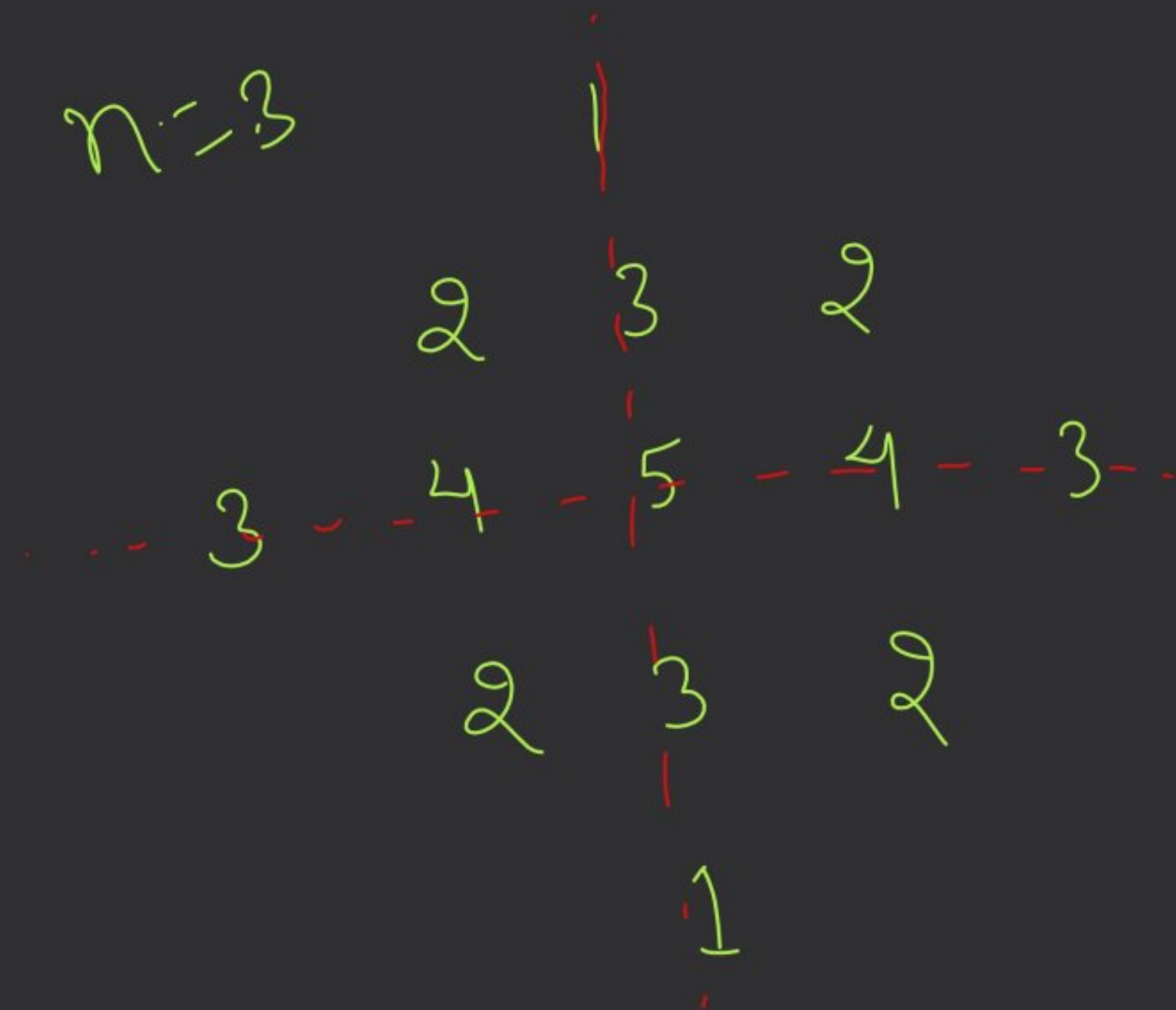
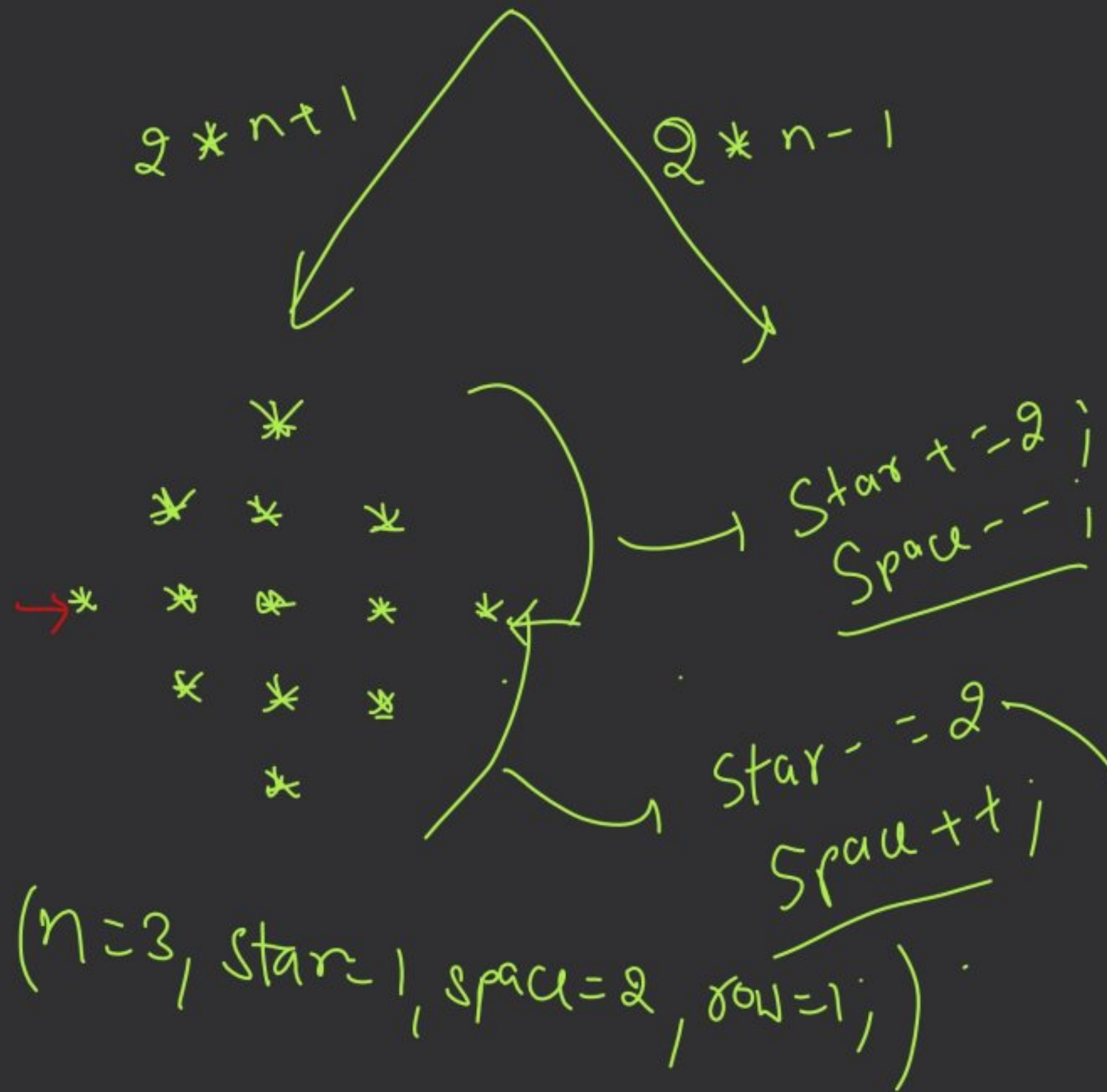


$n=3$



$n=3,$



While ($row \leq 2 * n - 1$) {

// Space

int i=1;

while ($i \leq space$) {

$cout << " "$;

$i++$;

}

$Star = Star - 2$;

// star

int j=1;

while ($j \leq star$) {

$cout << "*" << " "$;

$j++$;

}

if ($row \leq n$) {

$star+=2$;

$space--$;

} else {

$star-=2$;

$space++$;

}

3 4 5 4 3 // star

~~3~~ 4 5 4 3
2 3 2
1

n=3, space=n-1, row=1, star=1, val=1;

while (row <= 2 * n - 1) {

int i=1;

while (i <= space) {

cout << " ";

i++;

}

(space)

int j=1;

int number=val;

while (j <= star) {

cout << number;

if (j < $\frac{\text{star}}{2} + 1$) {

number++;

} else {

number--;

j++;

(j < 3)

if (row < n) {

star = star + 2;

space = space - 1;

val++;

} else {

star = star - 2;

space = space + 1;

val--;

cout << endl;

row++;

-13
7

Maths Concept →

(extraction of digit)

$N = 7789$

$(7, 7, 8, 9)$

$\downarrow$
 $\cdot 10$
 $9,$

778

$\swarrow$
 $\cdot 10$
 8

77

$\swarrow$
 $\cdot 10$
 7

$\swarrow$
 $\cdot 10$
 7

$10 \overline{) 710}$
 $\underline{0}$
 7
 $\underline{7}$
 0

Ques. 1. Sum of digit $\Rightarrow$

(5487)
int sum = 0;
while (n > 0) {

int digit = n % 10;

n = n / 10;
sum = sum + digit;
}

Dry Run		
Sum	n	digit
0	5487	7
7	548	8
15	54	4
19	5	5
<u>24</u>	0	

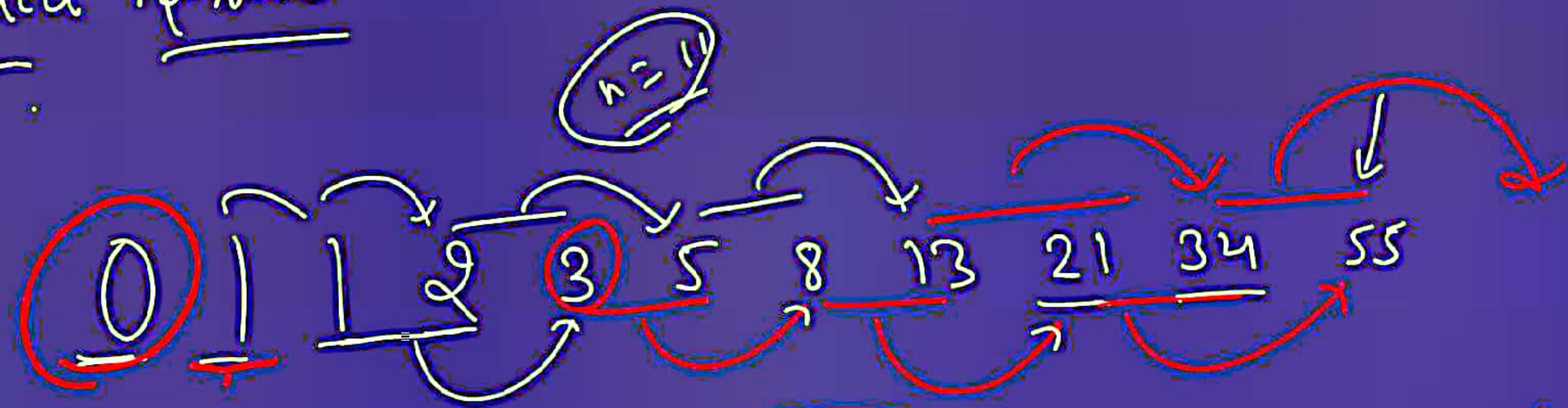
Sort (sum) -

H.W.
n = 587623189
 $\rightarrow$ Revise the today's class


```
File Edit Selection View Go Run Terminal Help
Pattern1.java Pattern2.java Star_Rhombus.java MirrorPattern.java pattern10.java Pattern_Rombus.java X
Pattern_Rombus.java > Pattern_Rombus > main(String[])
3 public class Pattern_Rombus {
4     public static void main(String[] args) {
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25         if(j < star/2 + 1){
26             number++;
27         }else number--; //vertical mirroing done
28         j++;
29         // number++;
30     }
31     //mirror
32     if(row < n){
33         star += 2;
34         space--;
35         val++; //extra addition
36     }else{
37         star -= 2;
38         space++;
39         val--; //extra addition
40     }
41     System.out.println();
42     row++;
43 }
44 sc.close();
45 }
46 }
47 }
```

```
1 import java.util.Scanner;
2
3 public class Pattern_Rombus {
    Run | Debug
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int row = 1, space = n-1, star = 1, val = 1;
8
9         while(row <= 2*n-1){
10             //space
11             int i = 1;
12             while(i <= space){
13                 System.out.print(s: "\t");
14                 i++;
15             }
16
17             //star
18             int j = 1;
19             // row start point is hidden on the val , save it on a number
20             int number = val;
21             while(j <= star){
22                 // System.out.print(val + "\t");
23                 System.out.print(number + "\t");
24
25                 if(j < star/2 + 1){
26                     number++;
```


Fibonacci Series



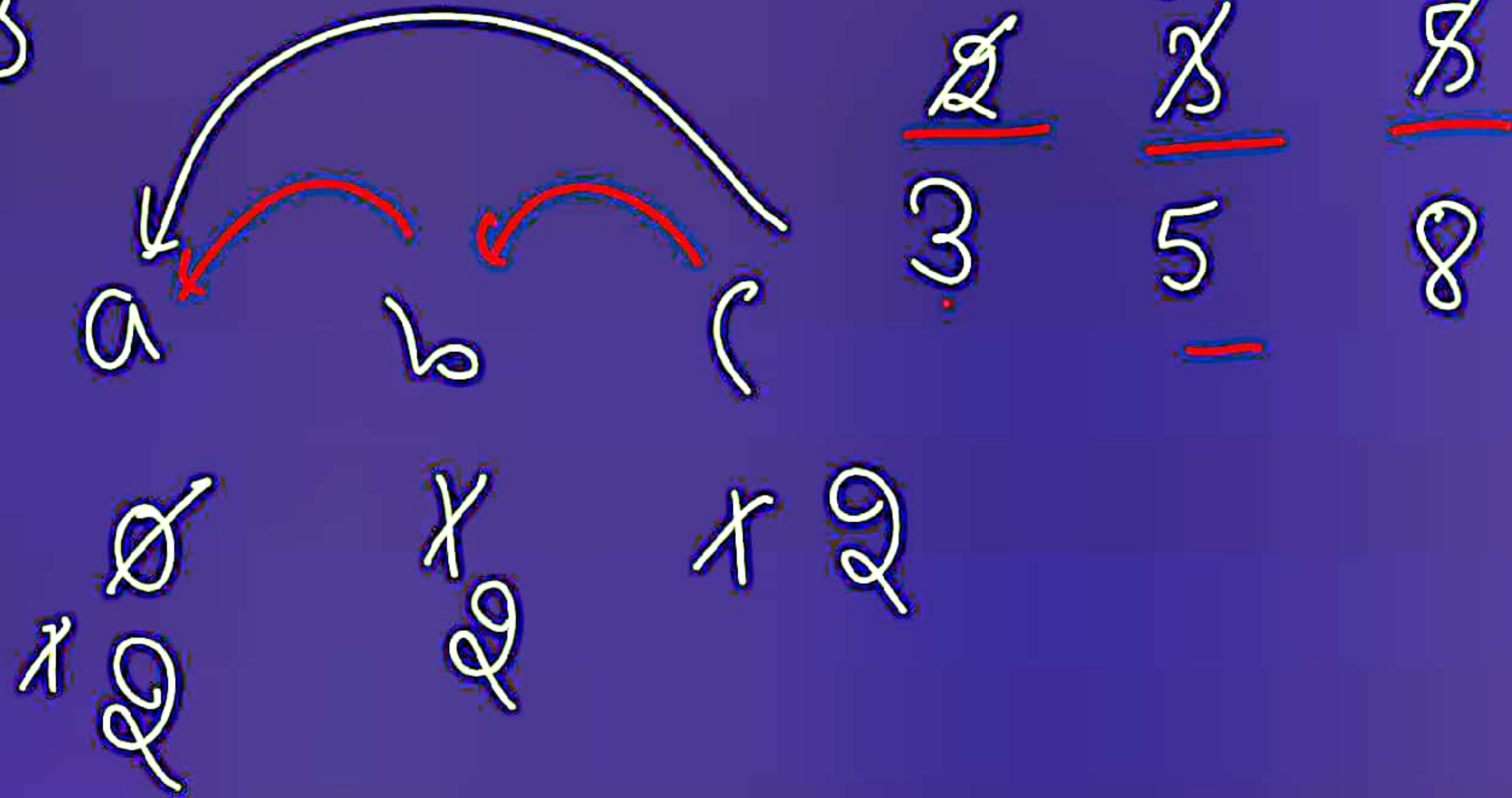
$$\text{Fib}(n) = \text{Fib}(n-1) + \text{Fib}(n-2);$$

$$\text{Fib}(11) = \text{Fib}(10) + \text{Fib}(9)$$

$$\begin{aligned} \text{Fib}(5) &= \text{Fib}(4) + \text{Fib}(3) \\ &= 2 + 1 \\ &= 3 \end{aligned}$$

0 1 1 2 3

$$a = b;$$



```
int a=0, b=1;
int n=11; int i=1;
```

```
while (i <= n) {
    System.out.print(a + " ");
```

```
    int c = a + b; ←
```

```
    a = b; ←
```

```
    b = c;
```

```
    i++;
```

```
}
```


$i = 1, 2, 3, 4, 5, 6, \dots, 11$

a	b	c
0	1	1
1	1	2
2	2	3
2	3	5
3	5	8
5	8	13

Output

0 1 1 2 3 5 8
13 21 34 55 89
144

Reverse of a number :-

int n = 5678, revnum = 0;

while (n > 0) {

int digit = n % 10;

n = n / 10;

revnum = revnum * 10 + digit;

}

5678

(8)

8765

1221 / ==
1221


```
if (n == reverseNo) {
```

```
    cout << "Palindrome";
```

```
else { cout << "not Palindrome";
```

```
}
```


Print all divisors :-

36 $\rightarrow$ 1, 2, 3, 4, 6, 9, 12, 18, 36

(1 2 3 4 5 6 7 8 9 9)

int orig = n;

int i = 1, n = 36;

while (i <= n) {

if (n % i == 0)

cout << i << " ";

if (i != n)

2³²

2147

if (i == n)

gcd

(20, 40)

(Common divisor)

$\text{Math.min}(a, b);$

$\text{min}(a, b)$

(16, 40)

(18, 40)

2

2 ✓

4 ✓

8 ✓


```
int a=16, b=40;  
int i=2, end = math.min(a,b);  
gcd=1;  
while(i <= end) {
```

```
if (a%i==0 && b%i==0) { ←
```

```
    gcd=i;
```

```
    i++;
```

```
}
```

Q17

(40, 80)

2

gcd = 2 / 4 / 16

2 / 40